NewsChain

A news app that starts from the outcome and rewinds time to reveal the full story.

Requirements Document

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1. Requirements Gathering

1.1 Purpose and Objectives for Requirements Gathering User Study

The objective of this user study was to understand how people currently consume news and to identify pain points that News Chain can address. Specifically, the study aimed to:

- Understand the news consumption habits and preferred platforms of the target users.
- Identify how often users feel confused or lacking context when encountering a breaking news headline.
- Determine user interest in a 'reverse-timeline' news format that starts with a headline outcome and progressively reveals background context.
- Assess user preferences for news depth, from brief summaries to full historical chains of events.
- Understand what motivates or discourages users from reading full news articles to discover design priorities.
- Evaluate whether users would want personalization features (e.g., topic preferences, saved story chains).

1.2 Choice of User Study and User Sample

User Study Type: An online survey was selected as the user study method. A survey allows for rapid, scalable data collection across a diverse group and is well-suited for gathering both quantitative data (frequency, ratings) using Google form.

Target Users: The target users for News Chain are adults across a wide age range who regularly consume digital news, particularly through smartphones. This includes:

- University and college students who need to stay informed on current events for academic or civic engagement.
- Young professionals who keep up with industry news but have limited time to read full articles.
- Casual news readers who feel overwhelmed or under-informed by conventional news feeds.

Dual Perspective: A single user type (general digital news readers) was used for this study, as News Chain is primarily a consumer-facing news application without a content-publisher side.

Sample Size: 12 participants were surveyed. The participants were housemate, friends and relatives.

1.3 Draft of the User Study Survey

The following survey was administered to participants online (Google Forms):

<u>Section A – Demographic Information</u>

- Q1. What is your age range?
  - Under 18 / 18–24 / 25–34 / 35–44 / 45 or older

- Q2. What is your occupation?
  - Student / Working Professional / Freelancer / Unemployed / Other
- Q3. How many hours per day do you typically spend consuming news?
  - Less than 30 min / 30 min–1 hour / 1–2 hours / More than 2 hours

Section B – Current News Habits

- Q4. Which platform(s) do you use most to follow news? (Select all that apply)
  - Social Media / News Apps (e.g., Apple News, Flipboard) / News Websites / TV/Radio / Word of mouth / Other
- Q5. On a scale of 1–5, how often do you feel you lack background context when you encounter a breaking news headline? (1 = Never, 5 = Always)
- Q6. When you encounter a confusing news story, what do you typically do?
  - Search online for more context / Ask someone / Ignore it / Skim multiple headlines / Other

Section C – Interest in NewsChain's Concept

- Q7. How interested are you in a news format where stories are presented starting from the latest headline and you can tap backwards to see how the story developed over time? (1 = Not interested, 5 = Very interested)
- Q8. How important is it that a news app gives you the full historical context of a story? (1 = Not important, 5 = Very important)
- Q9. Would you prefer news stories presented as: a reverse timeline / traditional chronological order / no preference
- Q10. On a scale of 1–5, how valuable would it be to see a visual 'chain' connecting past events to a current headline? (1 = Not valuable, 5 = Very valuable)

Section D – Feature Preferences

- Q11. Which features would you most want in a news app? (Select all that apply)
  - Bookmarking articles / Personalized topic feed / Offline reading / Story chain visualization / Share story chain / Dark mode / Audio summary
- Q12. How important is it for the app to be available offline? (1 = Not important, 5 = Essential)

1.4 Data Visualization and Analysis

The following tables summarize the quantitative findings from the 12-participant survey.

Q3 – Daily News Consumption Time

Time Spent	Responses	Percentage
Less than 30 min	3	25%
30 min – 1 hour	6	50%

1 – 2 hours	2	17%
More than 2 hours	1	8%

The majority of users (50%) spend 30 minutes to 1 hour per day on news consumption, indicating a moderate but consistent engagement level.

Q5 – Frequency of Lacking Context (Mean Score)

Metric	Score (out of 5)
Mean	3.8
Median	4
Mode	4

A mean score of 3.8 out of 5 indicates that users frequently feel they lack background context when reading breaking news. This strongly validates the core concept of NewsChain.

Q7 – Interest in Reverse-Timeline News Format

Rating	Responses	Percentage
1 – Not Interested	0	0%
2	1	8%
3 – Neutral	2	17%
4	4	33%
5 – Very Interested	5	42%

Mean: 4.1 / 5. 75% of respondents rated their interest as 4 or 5, indicating strong demand for the reverse-timeline concept that is central to NewsChain.

Q11 – Most Wanted App Features

Feature	Votes	%
Story chain visualization	10	83%
Personalized topic feed	9	75%
Bookmarking articles	8	67%
Dark mode	7	58%
Audio summary	5	42%

Offline reading	4	33%
Share story chain	4	33%

1.5 Insights from the Results

- Users frequently encounter news without sufficient context: A mean score of 3.8/5 on Q5 confirms that lack of background information is a widespread pain point, directly validating the NewsChain concept of presenting stories as historical chains.
- Strong interest in the reverse-timeline format: 75% of users rated their interest as 4 or 5, confirming the unique reverse-timeline format of NewsChain has genuine market appeal.
- Story chain visualization is the most desired feature: 83% of respondents selected it, reinforcing that the visual representation of a story's evolution is a key differentiator to build as a core feature.
- Personalization is highly valued: 75% want a personalized feed, meaning NewsChain should include topic preference settings to tailor which story chains appear for each user.
- Users are time-constrained: Most users spend 30–60 minutes per day on news, suggesting short, scannable chain summaries are essential before users dive into full chain exploration.

2. Requirements Generation

2.1 Personas and Scenarios

Persona 1 – Maya Chen, the Overwhelmed Student

Attribute	Detail
Name	Maya Chen
Age	21
Occupation	Science student at a large university
Tech use	Heavy smartphone user; uses Instagram, Reddit, and TikTok for news
Frustration	Constantly sees breaking headlines that reference events she doesn't know about. Feels embarrassed in class discussions when she lacks context.
Goal	To quickly understand the background of a news story without hours of research

Scenario: Maya opens her phone between classes and sees a headline: 'Parliament passes emergency AI regulation bill.' She has no idea what led to this. She opens NewsChain, taps the headline, and is immediately presented with the most recent event node. She swipes backwards through a visual chain: the earlier Senate hearings, the AI incident that triggered public debate, and the original research paper from two years ago that started it all. In under 5 minutes, Maya understands the full story. She saves the chain to review later for a class essay.

Persona 2 – David Tanaka, the Busy Professional

Attribute	Detail
Name	David Tanaka
Age	34
Occupation	Product Manager at a software company
Tech use	Uses Apple News and LinkedIn for news; checks phone during commute
Frustration	News articles are long but shallow. He wants the gist of a story's full history fast, without reading five separate articles.
Goal	Stay informed on tech and business news efficiently during his 20-minute commute

Scenario: David is on the SkyTrain heading to work. He sees that a major tech company just announced a massive layoff. He opens NewsChain and taps the story. He is shown a concise chain: today's layoff announcement, the earnings report from last quarter, the pivot strategy announced 6 months ago, and the acquisition that started everything 2 years prior. He skims each node summary and feels fully caught up by the time he arrives at his stop. He shares the story chain link with a colleague via the app's share feature.

Persona 3 – Sandra Kowalski, the Sceptical News Reader

Attribute	Detail
Name	Sandra Kowalski
Age	40
Occupation	High school teacher and parent of two
Tech use	Moderate smartphone user; uses Facebook and Google News
Frustration	Distrusts sensationalist headlines; wants to see which sources are cited and whether a story has been consistent over time.
Goal	Find trustworthy, well-sourced news she can confidently discuss or share with students

Scenario: Sandra reads a controversial headline about a new school policy being debated in her

province. She opens NewsChain and navigates the chain. She notices each node displays the source publication and a source-reliability badge. She can see that the story's framing changed significantly between outlets over the months, and she can tap through to read the original government document. Feeling confident in the sourced information, she decides to use the story chain as a classroom discussion resource, having bookmarked it the night before.

Persona 4 – Kenji Ohmura, the Casual News Scroller

Attribute	Detail
Name	Kenji Ohmura
Age	25
Occupation	Barista and part-time freelance photographer
Tech use	Heavy TikTok and Instagram user; occasionally reads news from social shares
Frustration	Finds traditional news apps boring and cluttered with too much text and unrelated ads.
Goal	Stay loosely informed on world events in a visually engaging, low-effort way

Scenario: Kenji is on his break at work and scrolls through NewsChain's home feed. The card-based layout with bold headline images catches his eye. He swipes through a visually-presented chain about a climate summit, appreciating the clean cards with icons representing each event node. He doesn't read every card fully but skims the visual timeline enough to feel informed. He taps the 'audio summary' button and listens to a 90-second recap of the full chain while cleaning equipment, completing his news catch-up effortlessly.

2.2 Requirements Generated from Persona and Scenario Analysis

The following requirements were derived by analyzing each persona's goals, frustrations, and scenarios:

From Maya Chen (Persona 1):

- The app must present a visual reverse-timeline chain for each news story.
- Each node in the chain must include a short summary (2–3 sentences) viewable without opening the full article.
- Users must be able to bookmark or save story chains for later reference.
- The app must allow users to swipe or tap backward through a chain naturally on mobile.

From David Tanaka (Persona 2):

- Chain node summaries must be scannable in under 30 seconds each.
- The app must provide a 'full chain summary' view that gives an overview of the entire story arc in one scroll.
- A share feature must allow users to send a story chain link to others (via messaging apps,

email, etc.).
- The app must function well in low-connectivity environments (e.g., during transit).

From Sandra Kowalski (Persona 3):

- Each node must display the source publication name and date clearly.
- The app should include a source credibility or reliability indicator on each node.
- Users must be able to tap through to read the original full article from any node.
- Bookmarks must be organized and persistently saved across app sessions.

From Kenji Ohmura (Persona 4):

- The home feed must use a card-based visual layout with compelling imagery.
- The app must support a dark mode for comfortable viewing in different lighting.
- An audio summary feature should allow users to listen to a 60–90 second recap of any story chain.
- The UI must be clean and uncluttered, with no intrusive advertisements interrupting story flow.
- Chain nodes should include visual icons or category tags (e.g., Politics, Tech, Climate) for quick orientation.

3. Synthesis of Requirements Gathering and Generation

3.1 Prioritized Requirements List

The following requirements combine insights from the user study and the persona-based requirements generation. They are categorized as Must-Have (high priority) and Want-to-Have (lower priority).

Must-Have Requirements (High Priority)

- Reverse-timeline story chain: The app must display news stories as a chain of linked events ordered from the most recent outcome backwards to the origin. This is the core differentiating feature of NewsChain.
- Node-based story display: Each event in the chain must be displayed as a distinct card/node with a headline, source, date, and a brief 2–3 sentence summary.
- Swipe-back navigation: Users must be able to navigate backwards through the chain using intuitive swipe or tap gestures, consistent with mobile UX conventions.
- Source attribution: Every node must clearly display the original news source name and publication date.
- Link to full article: Users must be able to tap any node to open the original full article, either in-app or in a browser.
- Bookmarking: Users must be able to save story chains to a personal reading list for future access.
- Home feed: A visually-engaging, card-based home screen must display trending and personalized story chains.
- Onboarding and topic selection: On first launch, users must be able to select their preferred news topics to personalize their feed.

Want-to-Have Requirements (Medium Priority)

- Full chain summary view: A dedicated single-screen view that summarizes the entire story arc in one pass, ideal for time-constrained users like David.
- Source credibility indicator: A visual badge or rating alongside each source to help users like Sandra assess reliability.
- Dark mode: A full dark mode theme for the app interface to support comfortable reading at night or in low-light settings.
- Category and topic tags on nodes: Visual tags (e.g., 'Economy', 'Climate', 'Tech') to help users orient themselves within a chain.

Low Priority / Nice-to-Have Requirements

- Offline reading: Allow users to download story chains for offline reading during low-connectivity situations.
- Push notifications: Smart, minimal notifications for major developments in a user's followed story chains.
- Search functionality: Allow users to search for specific stories, topics, or keywords to find relevant chains.
- Chain comparison: Allow users to view how different outlets frame the same chain of

events side by side.
- Audio summary: Users can listen to news without using their hands.

3.2 Conclusions from the Requirements Process

- The core News Chain concept of reverse-timeline news is strongly validated by both user study data and persona analysis. It directly addresses the near-universal pain point of lacking context for breaking news.
- Visual design and UX simplicity are critical: Cluttered UI is the most common reason users abandon news apps.
- Source credibility and trust are underserved in current news apps, and building transparency into each chain node is a meaningful differentiator that will attract more skeptical, educated users.

4. AI Use Section

4.1 Table of AI Tools and Specific Use

AI Tool Name	Version / Account Type	Specific Feature Used
ChatGPT	GPT-4o mini, Free account	Showing example of user requirement and checking the quality of questionnaire
Claude (Anthropic)	Claude Sonnet 4.6, claude.ai Free	Formatting word file based on questionairee

4.2 Value Addition

- Adding exact example for making questions simpler and more understandable
- Validating questionnaire against the prototype: All personas, scenarios, and requirements were cross-checked against the actual Figma screens to ensure accuracy and applicability.
- Polish the document

4.3 Appendix: Prompt History

Prompt 1 – ChatGPT (March 02, 2026)

Prompt: can you tell me where I should improve my questionnaire as user requirement

Prompt 2 – ChatGPT (March 02, 2026)

Prompt: can you tell me example of questionnaire based on the application concept

Prompt 3 – ChatGPT (March 09, 2026)

Prompt: what is main challenging point from user requirement

Prompt 4 – Claude (March 09 2026)

Prompt: Can you format word file based on my questionnaire results?

5. Work Date / Hours Log

Student: Sho Ogawa (300411060)

Date	Hours	Description of Work Done
Mar 2, 2026	1.0	Generated requirements list from personas; organized into Must-Have / Want-to-Have categories based on Figma prototype
Mar 4, 2026	1.0	Posting google questionnaire and talking about this to my housemate and friends
Mar 9, 2026	1.5	Wrote synthesis section, conclusions, and AI use section; cross-checked requirements against Figma prototype

Total Hours: approx. 3.5 hours

6. References

- Figma. (2024). Figma: The collaborative interface design tool. https://www.figma.com
- OpenAI. (2024). ChatGPT. https://chat.openai.com
- Anthropic. (2024). Claude AI. https://claude.ai